TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	52.1 ± 3.45
Correlation Stability	85.1 ± 1.51
Distribution Consistency	50.5 ± 9.29
Entropy Preservation	51.3 ± 2.10
Mean Z Score Drift	0.3 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	98.7 ± 1.85
Feature Accuracy	85.0 ± 1.10
Novelty Score	0.8 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: hepatitis.csv

1. Dataset Portrait

Attribute	Value
Rows	155
Columns	20
Numeric features	20
Categorical features	0
Missing cells (%)	5.4%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	41.200	36.823	-4.377	10.6%	Warn
Sex	1.103	1.000	-0.103	9.4%	OK
Steroid	1.506	1.306	-0.200	13.3%	Warn
Antivirals	1.845	1.960	+0.115	6.2%	OK
Fatigue	1.351	1.161	-0.189	14.0%	Warn
Malaise	1.604	1.710	+0.106	6.6%	OK
Anorexia	1.792	1.984	+0.192	10.7%	Warn
Liver Big	1.828	2.000	+0.172	9.4%	OK
Liver Firm	1.583	1.734	+0.151	9.5%	OK
Spleen Palpable	1.800	1.984	+0.184	10.2%	Warn
Spiders	1.660	1.863	+0.203	12.2%	Warn
Ascites	1.867	1.992	+0.125	6.7%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	52.1 ± 3.45
Correlation Stability (*2)	85.1 ± 1.51
Distribution Consistency (*3)	50.5 ± 9.29
Entropy Preservation (*4)	51.3 ± 2.10
Mean Z Score Drift (*5)	0.3 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	98.7 ± 1.85
Feature Accuracy (*10)	85.0 ± 1.10
Novelty Score (*11)	0.8 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.867741935483871	0.829032258064516	0.8677419354838708	-1.1102230246251566e-17	
F1	0.5933333333333333	0.2832967032967033	0.5800449550449549	-0.013288378288378283	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.7903225806451613	0.8387096774193548	0	
F1	0.5132478632478632	0.3159621424327307	0.464253850724439	-0.048994012523424293	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7806451612903226	0.8064516129032258	0.8	0.019354838709677413	
F1	N/A	0.0	N/A	0	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7903225806451613	0.8258064516129032	0.7935483870967742	0.003225806451612915	
F1	0.561449871078354	0.4240781440781441	0.5855310747193908	0.024081203641036737	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8516129032258064	0.7838709677419355	0.8580645161290322	0.006451612903225823	
F1	0.5385593354014405	0.2323853923853923 8	0.5759008827429879	0.03734154734154733	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.796774193548387 1	0.767741935483871	0.787096774193548 4	-0.00967741935483872 4	
F1	0.481414141414141 4	0.404026996395417 45	0.445845820845820 8	-0.03556832056832056	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.861290322580645	0.796774193548387 2	0.825806451612903 2	-0.03548387096774193 6	
F1	0.606257510605336 7	0.310084301663249	0.533540669856459 3	-0.07271684074887738	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.806451612903225 8	0.783870967741935 5	0.816129032258064 4	0.00967741935483872 4	
F1	0.366839826839826 8	0.136904761904761 92	0.385230651701239 93	0.01839082486141308	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.838709677419354 8	0.8	0.854838709677419 3	0.01612903225806452 3	
F1	0.571839213418160 8	0.326313131313131 3	0.535303030303030 4	-0.03653618311513047 4	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.806451612903225 8	0.806451612903225 8	0.806451612903225 8	0	
F1	N/A	0.0	N/A	0	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
2	50.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0	2.0
3	78.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0
5	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
8	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
9	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
10	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
11	39.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0
12	32.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
13	41.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
14	30.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
15	47.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
16	38.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
17	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
18	40.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
19	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
20	38.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	39.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	36.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	29.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	33.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
5	41.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	42.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
7	61.0	1.0	1.0	2.0	1.0	1.0	2.0	nan	nan	2.0
8	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
9	30.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
10	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	49.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
2	50.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
3	53.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
4	24.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
5	39.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
6	25.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
7	60.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	1.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
9	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	36.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	42.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	58.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
3	26.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
4	49.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
5	20.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	13.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
7	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	42.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	50.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	34.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
3	43.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	54.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
5	54.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	57.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
7	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
9	25.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
10	38.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	38.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
2	42.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
3	40.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	47.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
5	31.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	62.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
7	30.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	45.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	36.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
2	39.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
3	42.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
4	51.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
5	44.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	67.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
7	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
8	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
9	51.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
10	40.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	43.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	29.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	55.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
4	42.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
5	23.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
6	46.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
7	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
9	30.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	23.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	38.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	20.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
5	48.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
6	41.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
7	65.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
8	36.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
9	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
10	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	44.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	58.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	60.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	32.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
5	37.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	45.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0
7	38.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	44.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
9	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	28.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	35.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	28.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
3	34.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	24.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
5	43.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	56.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	28.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	49.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0